PROBLEM

Problem Statement 1 – System Decomposition & Analysis

Analyze an existing AI-based system by reverse engineering its architecture. Identify its major components, modules, data flow, and underlying AI algorithms, and explain how these components interact to provide the required functionality.

Problem Statement 2 – Architecture Reconstruction

Given a black-box software system, reverse engineer and reconstruct its overall system architecture. Identify the major modules and explain how they communicate and interact with each other using suitable architecture and component diagrams.

Problem Statement 3 – Algorithm Identification

Reverse engineer an existing machine learning application to identify its AI/ML algorithms, data preprocessing steps, training and prediction processes, and decision-making logic used to generate system outputs.

Problem Statement 4 – Data Flow & Process Mapping

Examine an existing distributed AI application and reverse engineer its data flow. Map the input data sources, processing pipeline, storage mechanisms, communication between components, and flow of information from input to final output.

Problem Statement 5 – Improvement & Redesign

Perform reverse engineering on an existing intelligent system to identify its limitations, scalability and performance issues. Propose an improved architecture and redesign strategy that enhances **efficiency, scalability, resource utilization, and response time**.

System used in your document: Google Maps – AI-Based Navigation and Route Planning System. The document identifies its GPS, map data, traffic analysis, route planning, and navigation modules.